

PT Javadwipa Duta Mandiri

September - December 2025

BMS: Energy Monitoring

(Building Management System: 3 Phase Energy Monitoring)

Defender Artha Widiprasetyo

5024221032

Pranaditya Tri Jyotista Vavitram P.S.

5024221052

Vania Bunga Febrina

5024221069

Internship Presentation

javadwipa



Company Profile

PT Javadwipa Duta Mandiri is a private company based in Surabaya that continues to grow in the field of innovative technology. The company focuses on Internet of Things (IoT)-based solutions to support the implementation of Industry 4.0 in Indonesia.

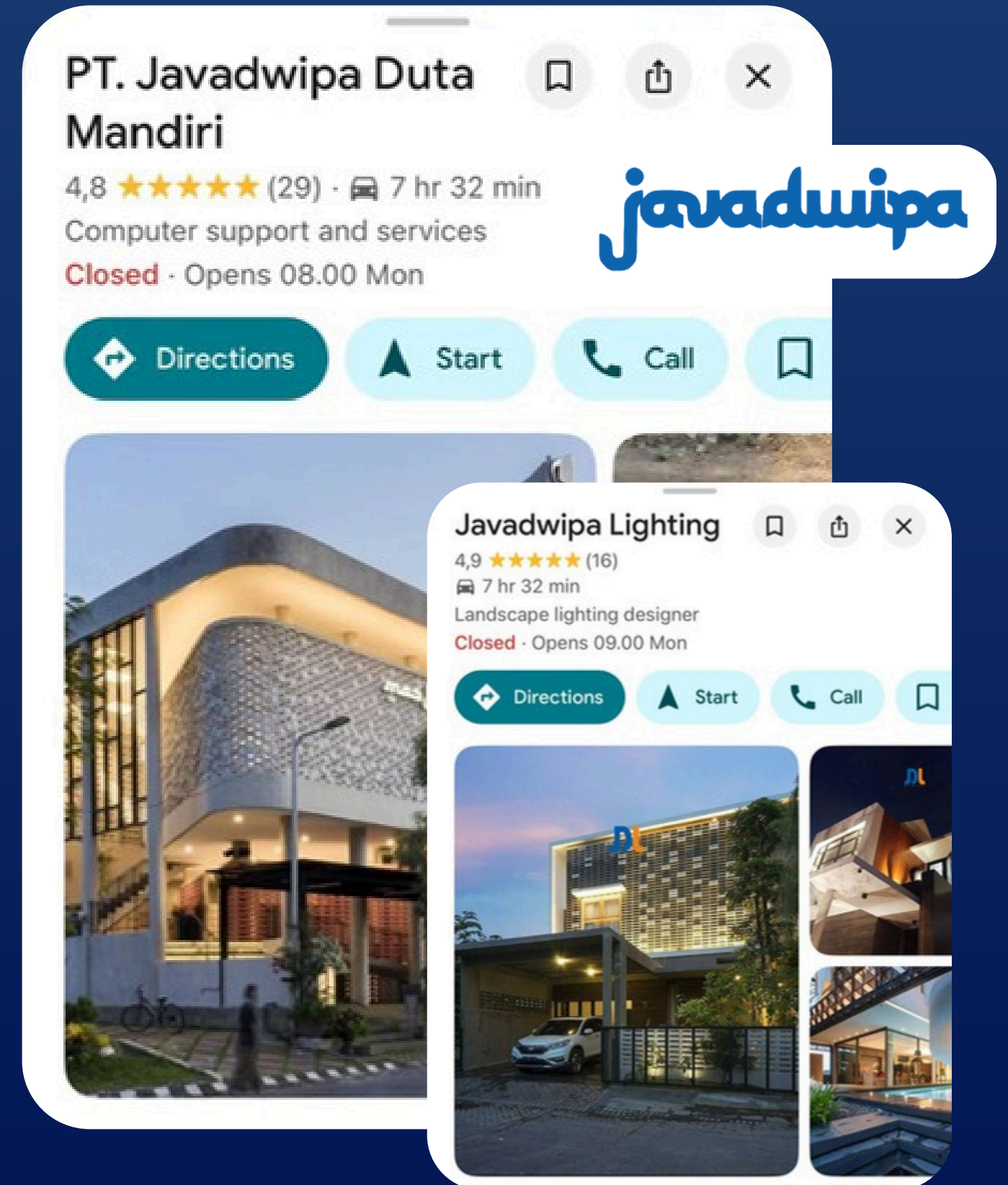
Its services cover a wide range of technology areas, including information and communication technology (ICT), network systems and security, smart lighting solutions, and integrated IoT-based system development.



Ruko Surya Inti Permata Blok C-7 60236 Kota Surabaya



www.javadwipa.com



Background

Why Energy Monitoring Systems Matter?



GLOBAL ENERGY AND CLIMATE CHALLENGES

This project aligns with SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action), focusing on energy efficiency and carbon reduction in buildings and industry.



THE NEED FOR SMART ENERGY MANAGEMENT

In the context of global energy crises and net-zero emission targets, intelligent and data-driven energy management has become essential, especially in industrial environments.



LIMITED VISIBILITY OF ELECTRICITY CONSUMPTION

Many organizations still lack real-time and accurate monitoring, leading to energy waste, higher operational costs, and inefficient resource use.



THE ROLE OF INTEGRATED MONITORING SYSTEMS

Real-time energy monitoring enables organizations to identify inefficiencies, support ISO 50001 implementation, and continuously improve energy performance.

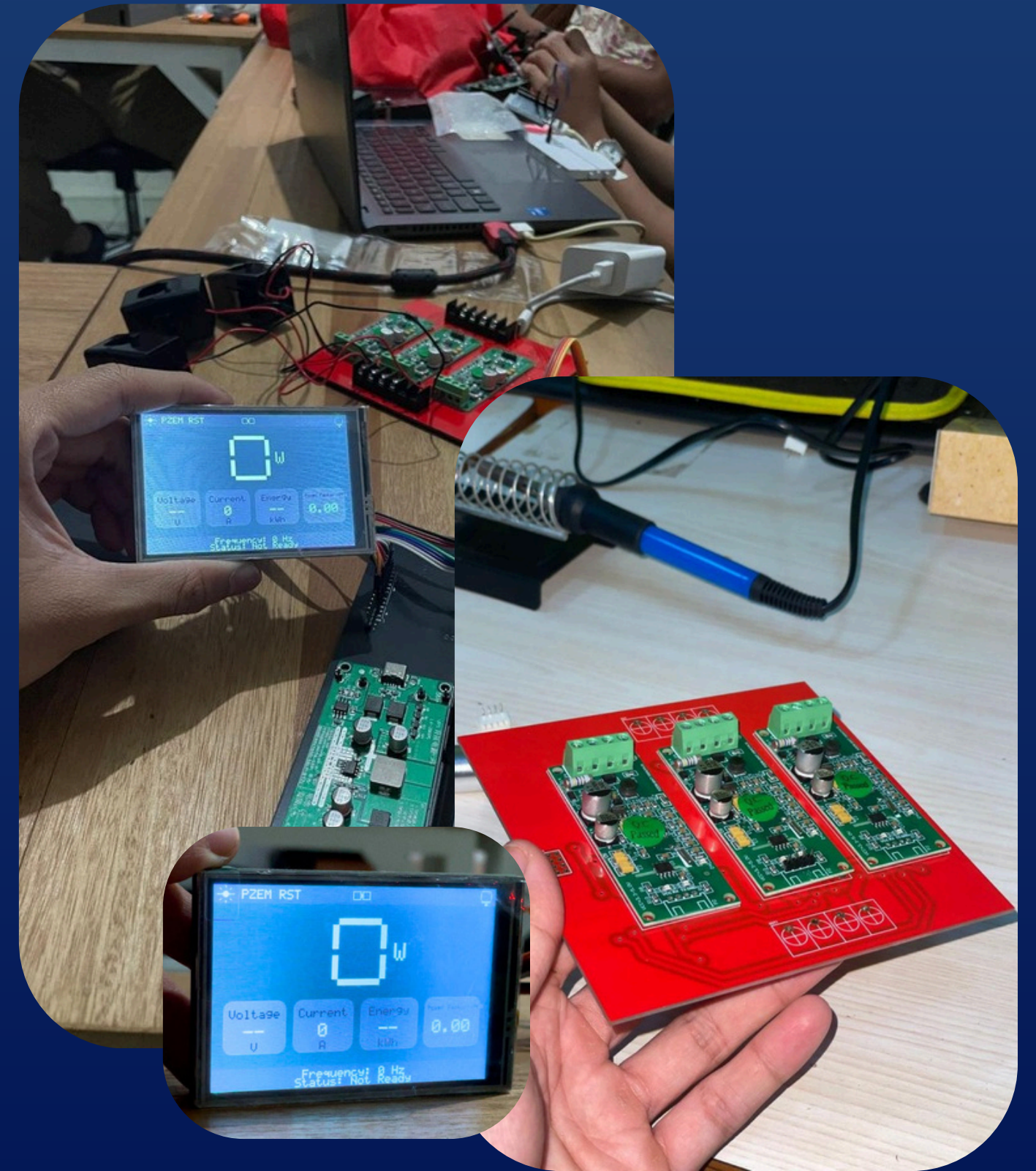


Internship Project

Building Management System: 3 Phase Energy Monitoring

During the internship, the main project focused on the development of a Building Management System (BMS) for energy monitoring.

The system was designed to monitor electricity consumption in both single-phase and three-phase electrical systems.



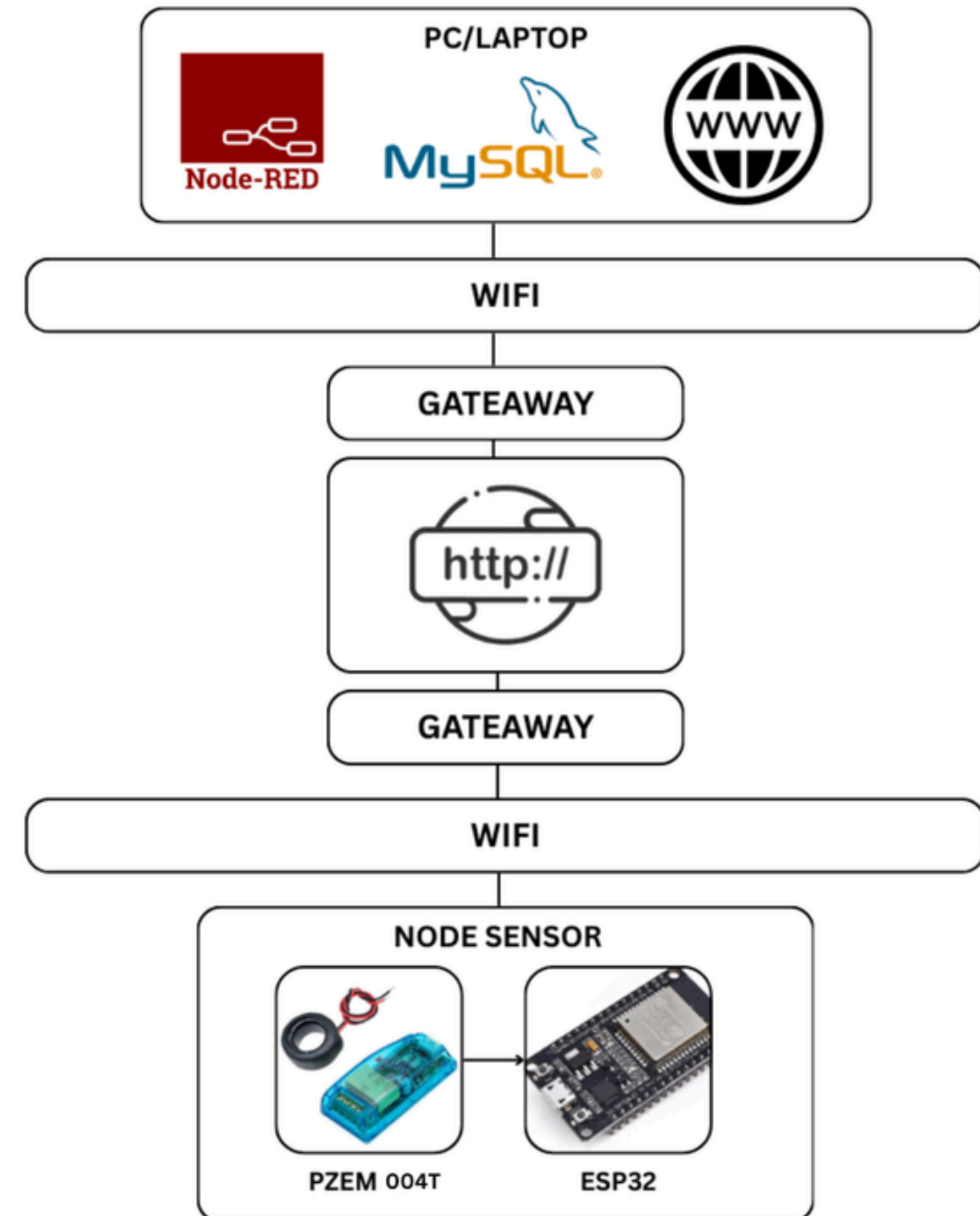
System Architecture

Building Management System: 3 Phase Energy Monitoring

The system starts with energy sensors and a microcontroller that collect electrical parameters directly from power lines.

The data is then transmitted wirelessly via Wi-Fi using HTTP communication.

Finally, the information is processed, stored, and visualized on a dashboard for real-time monitoring and analysis.

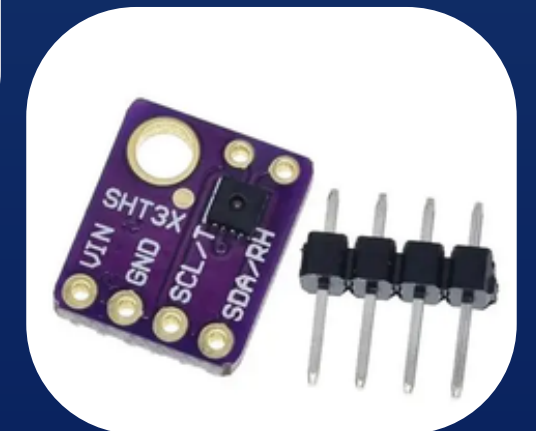
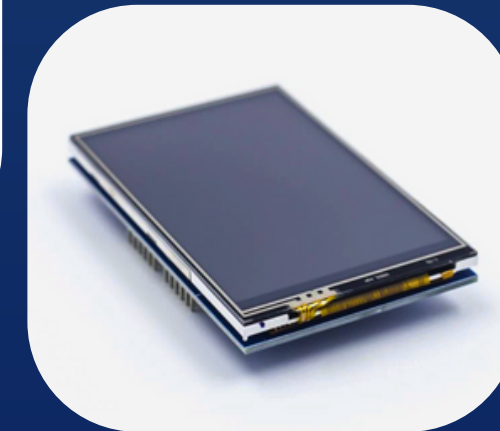


Technology Stack

Building Management System: 3 Phase Energy Monitoring

The system combines three PZEM-004T modules and an ESP32, integrating multiple UART sensors through RX/TX configuration and/or serial addressing.

A power supply converts 220V AC to 5V DC, with separation between high-voltage and low-voltage lines for safety and signal stability.



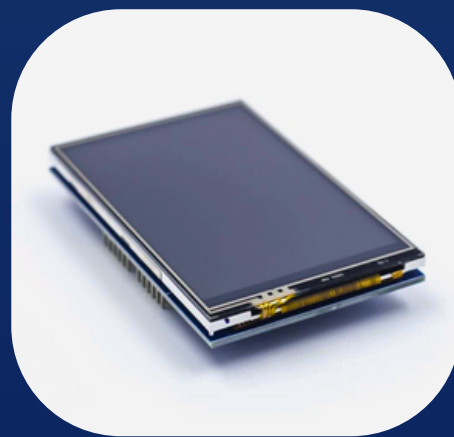
Technology Stack

Building Management System: 3 Phase Energy Monitoring



ESP32-S3

Acts as the main controller for data processing and Wi-Fi communication in the IoT system.



ILI9486 3.5inch

Displays system status and real-time measurement data locally.



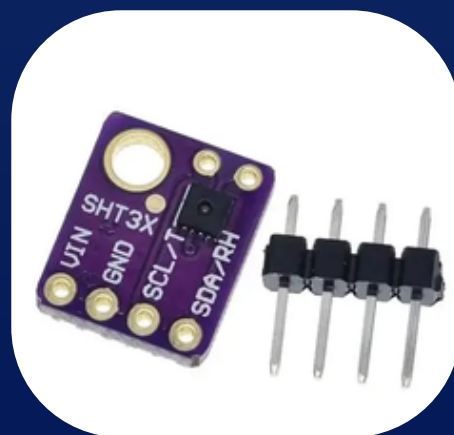
HLK-20M05

Converts 220V AC to a stable 5V DC supply to power the BMS system safely and reliably.



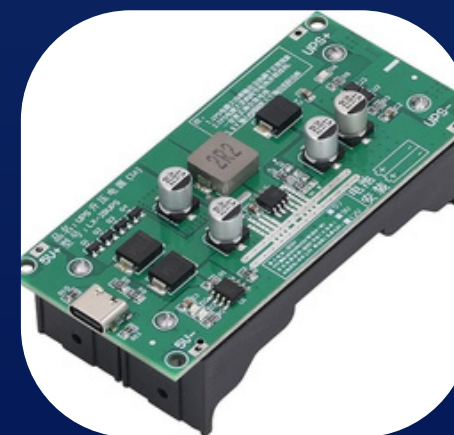
PZEM-004T

Measures voltage, current, power, energy, frequency, and power factor.



SHT31

Monitors temperature inside the BMS enclosure to ensure system safety.



UPS Module

Provides backup power to maintain system operation during power interruptions.

Internship Activities



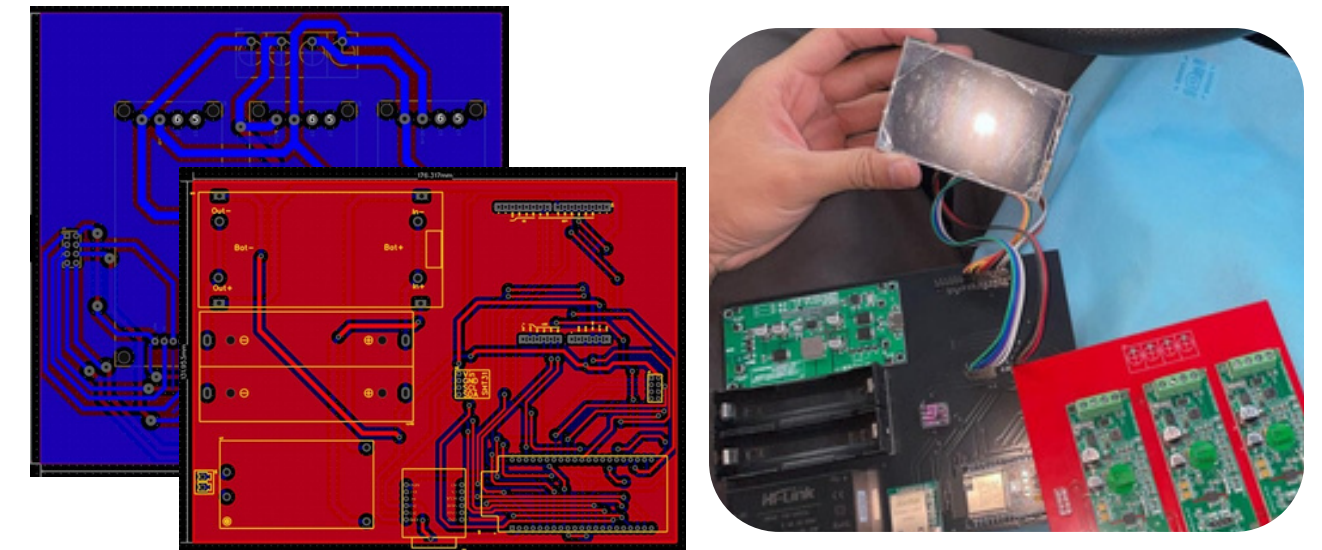
1) Research (Pre-Development)

We conducted field observation and technical discussions to validate the initial concept by reviewing an existing IoT-based energy monitoring system. This phase helped us define system requirements.



2) Implementation (Design & Build)

We designed the system architecture and electrical schematics, then built prototypes on a breadboard before moving to the final PCB design and fabrication. The hardware was assembled and packaged with clear separation between high-voltage and low-voltage sections

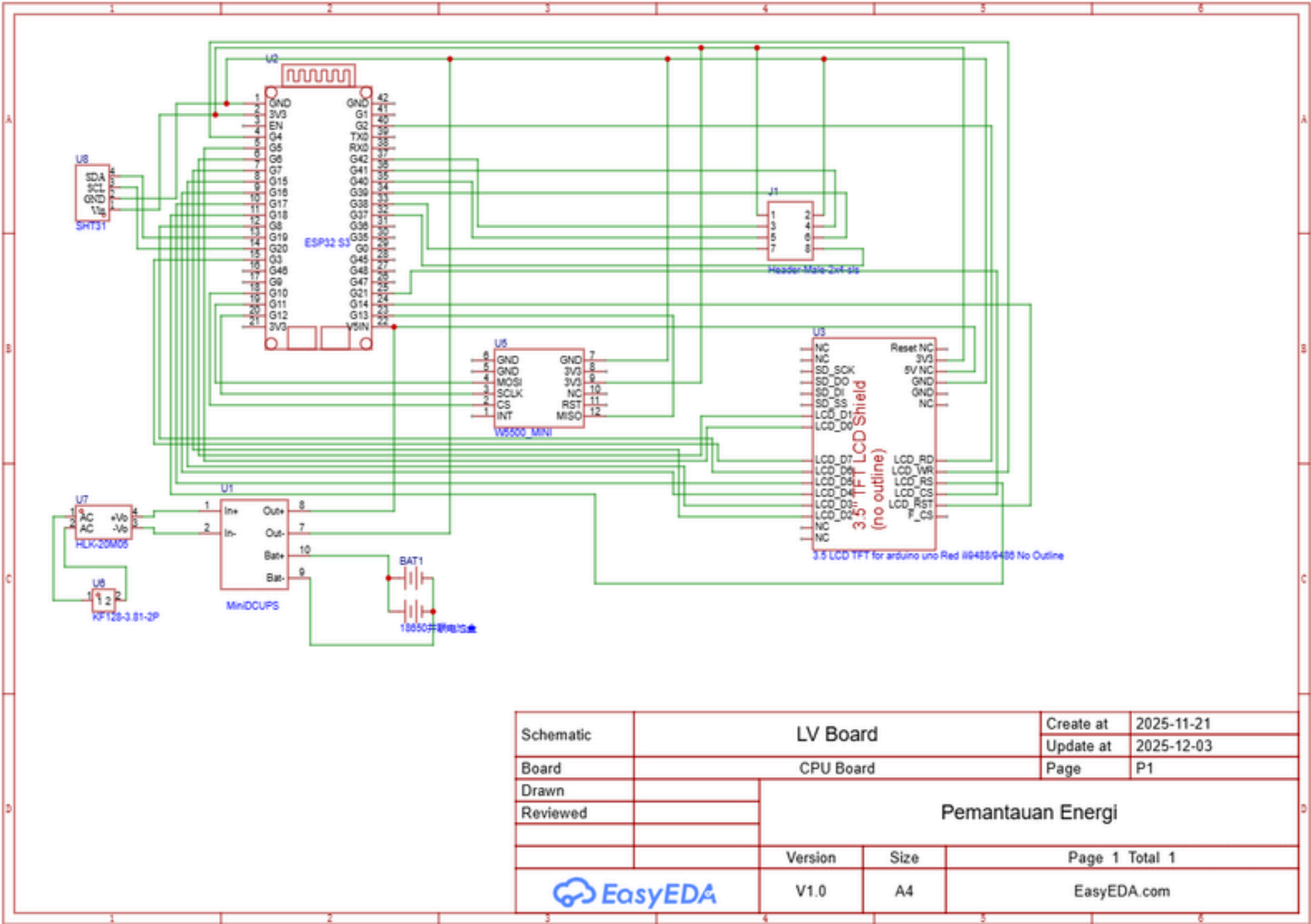
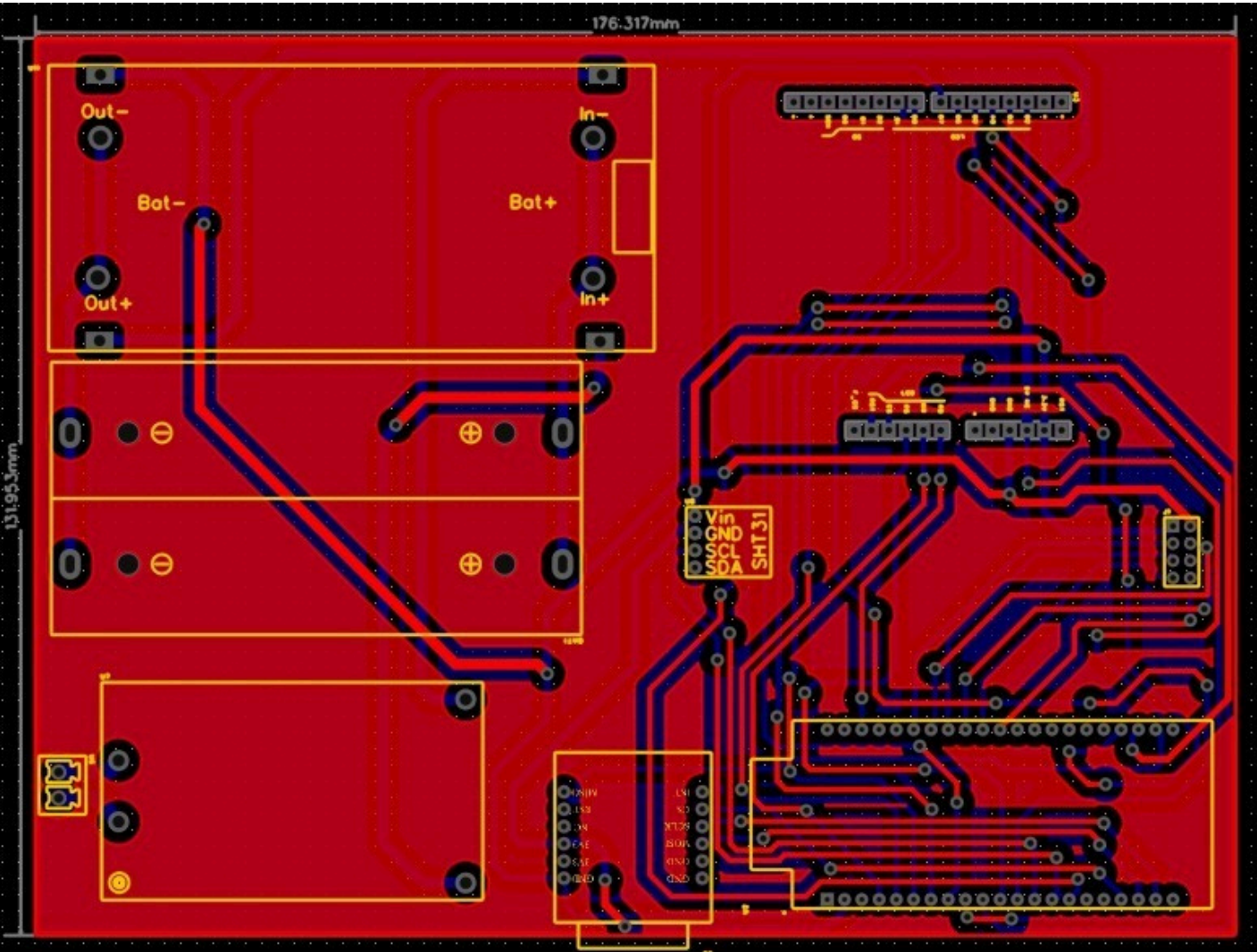


3) Testing (Trial & Verification)

We tested both single-phase and three-phase setups to verify that all key electrical parameters were read correctly. The results confirmed stable readings (voltage, current, power, energy, frequency, and power factor) and successful data delivery for monitoring purposes.

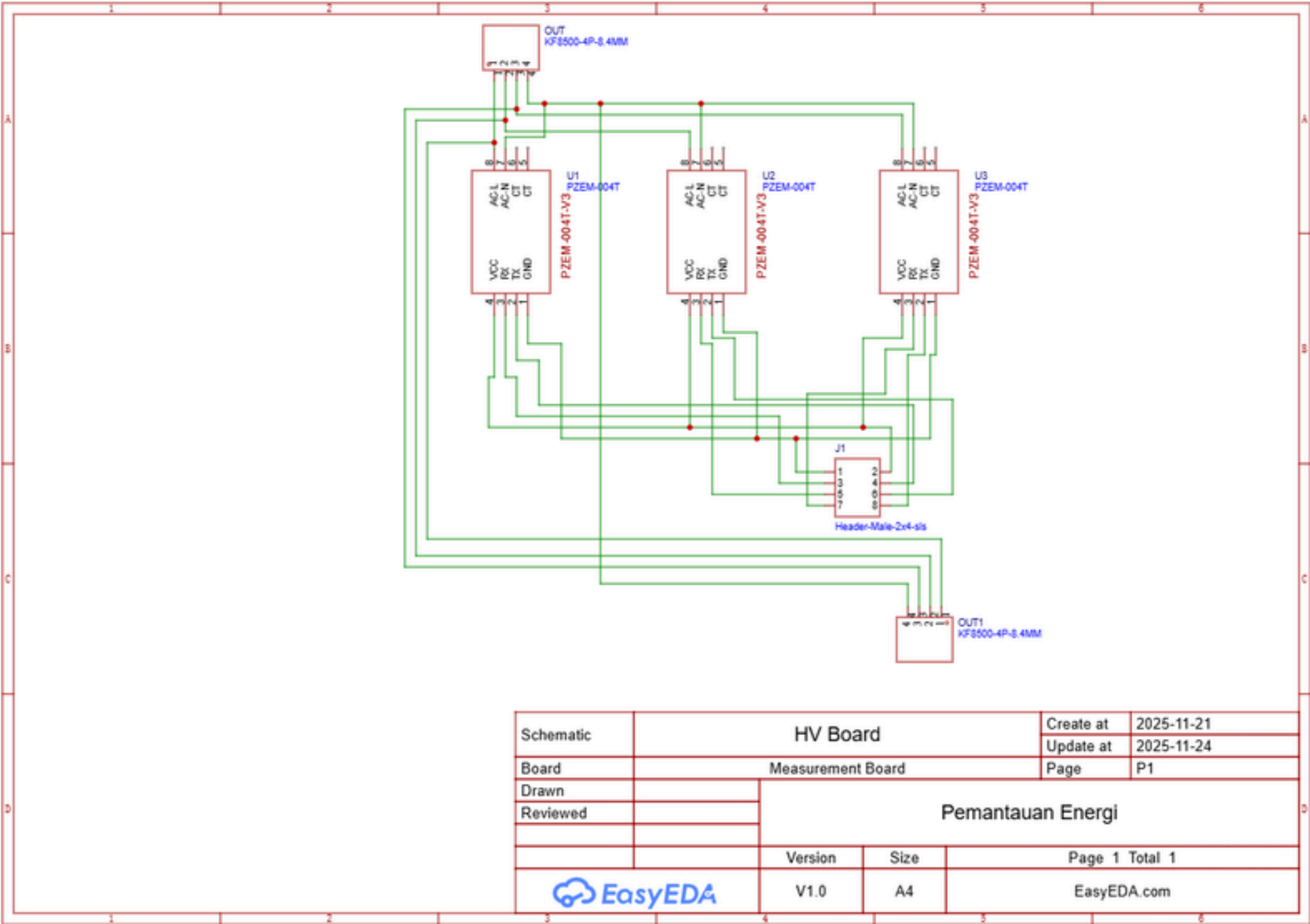
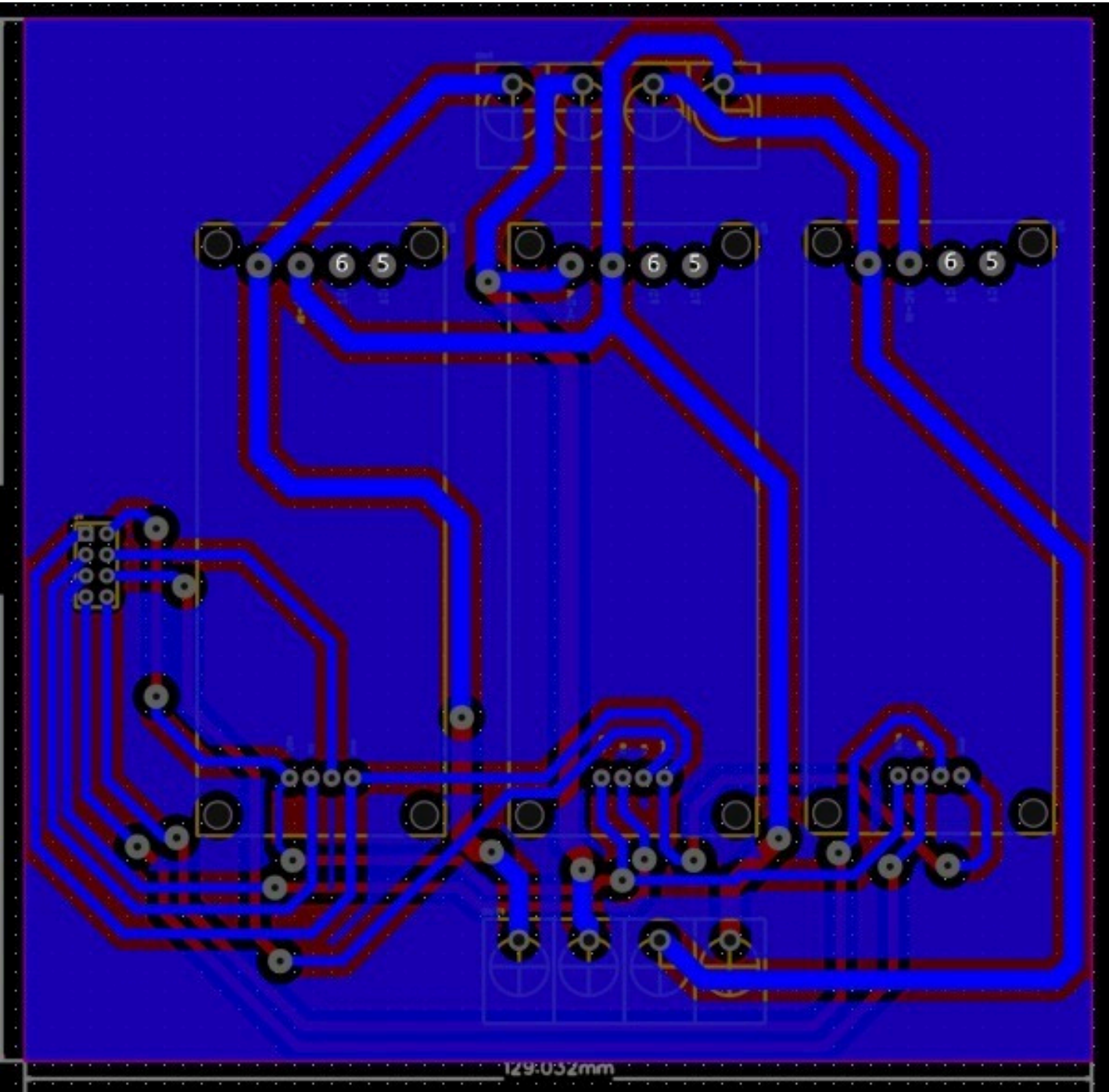
Technical Documentation

Low Voltage PCB Design



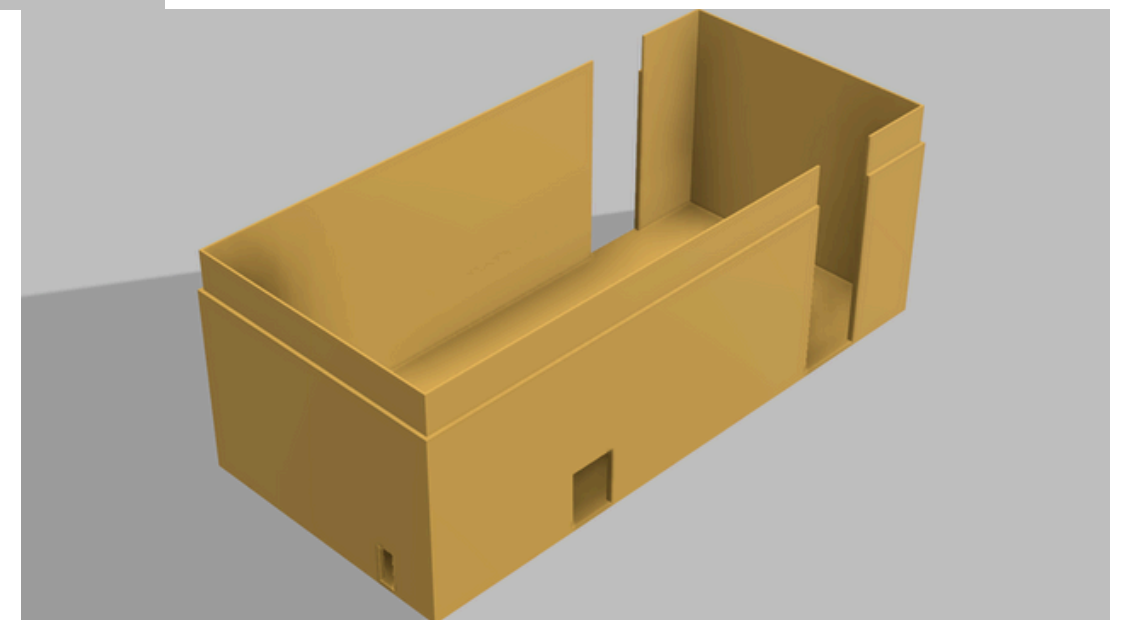
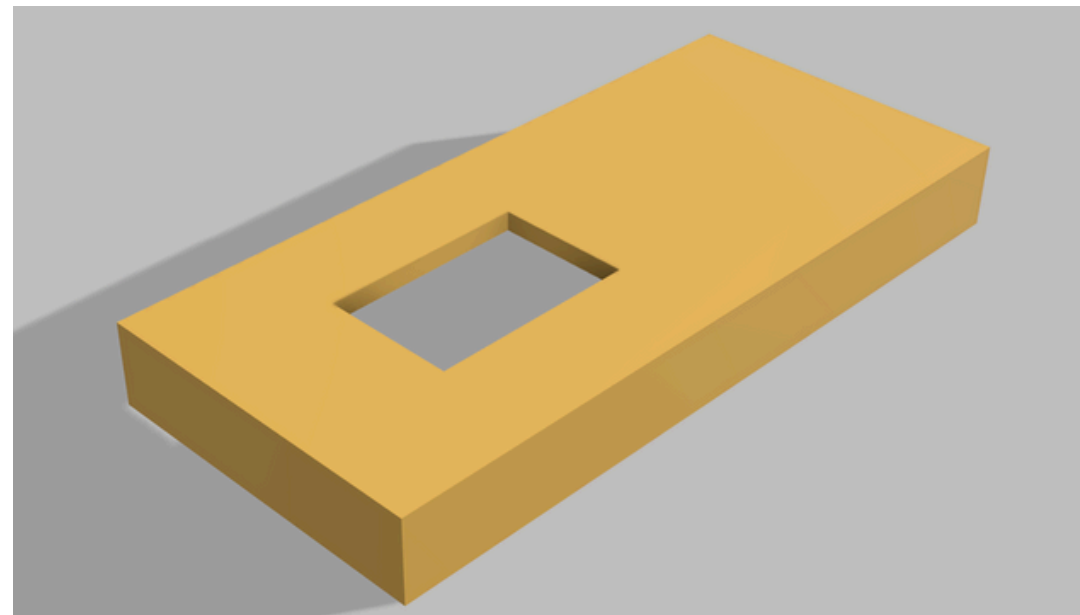
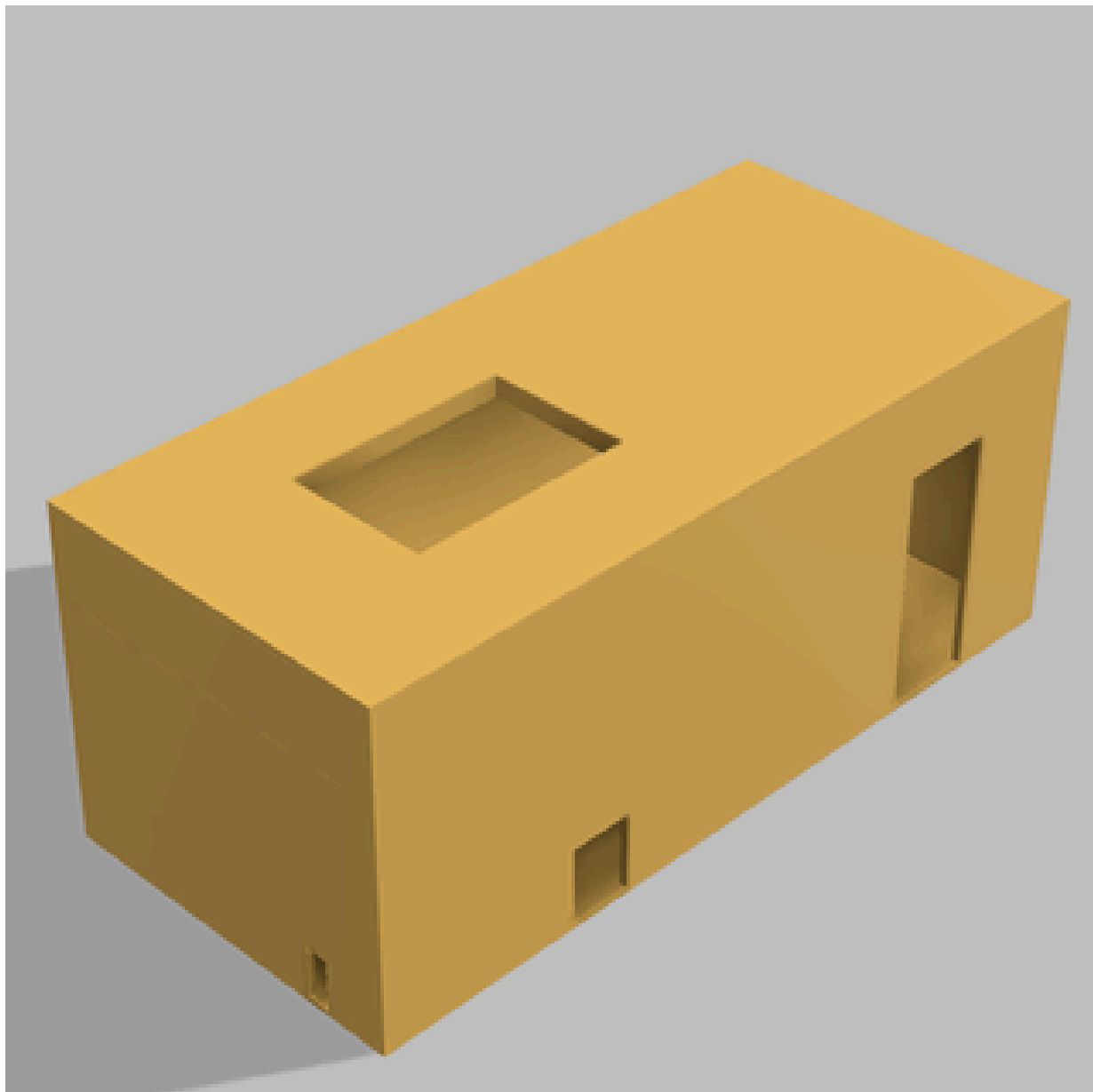
Technical Documentation

High Voltage PCB Design



Technical Documentation

3D Design of Casing

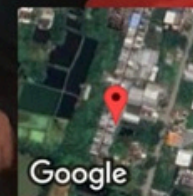
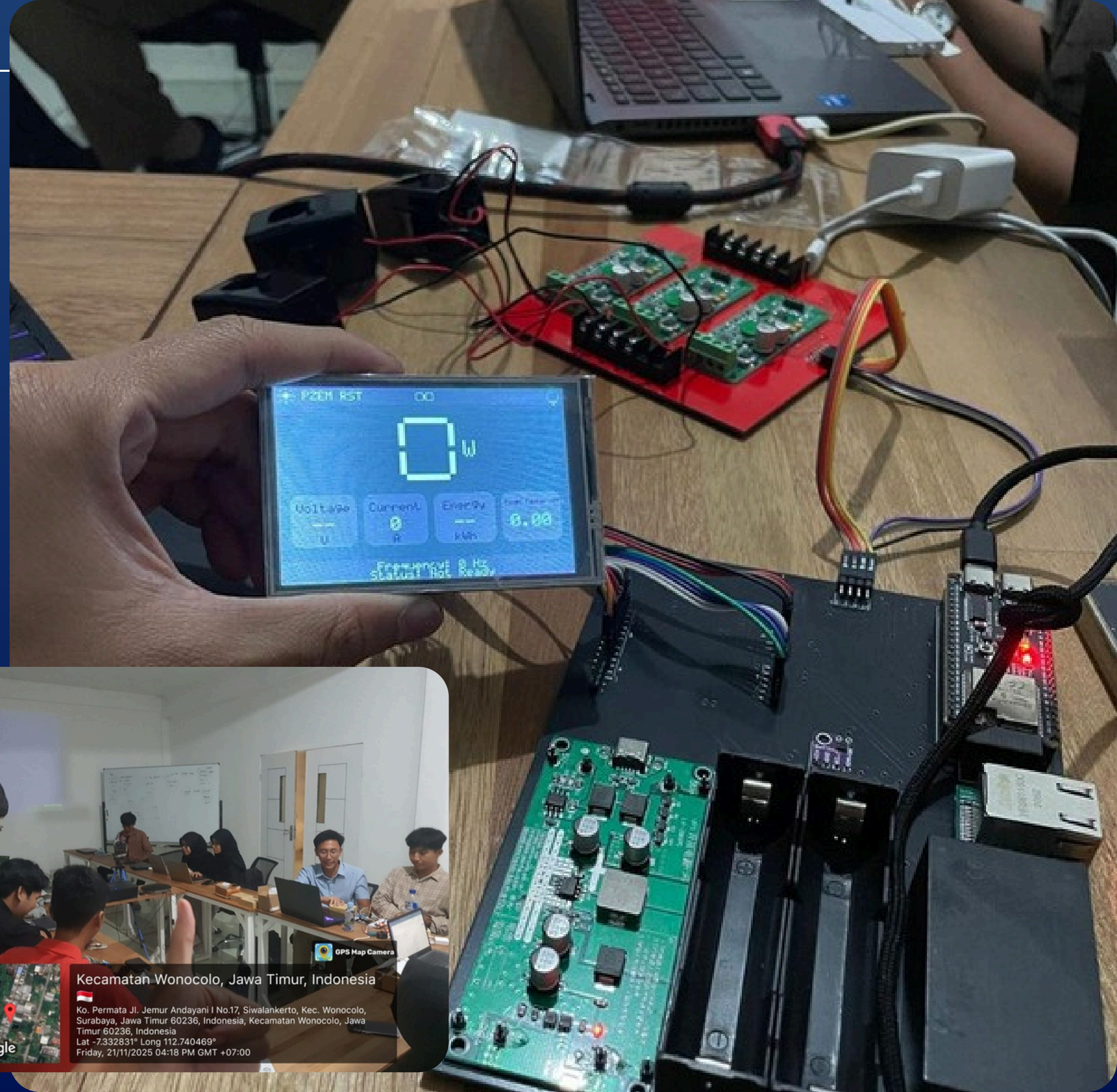


Results & Conclusion

A functional IoT-based energy monitoring system was successfully developed and implemented.

The system delivers stable, real-time electrical data for both single-phase and three-phase monitoring, supporting efficient and reliable energy management.

Overall, this internship provided valuable technical experience and practical insight into real-world industrial system development.



Kecamatan Wonocolo, Jawa Timur, Indonesia

Ko. Permata Jl. Jemur Andayani I No.17, Siwalankerto, Kec. Wonocolo, Surabaya, Jawa Timur 60236, Indonesia, Kecamatan Wonocolo, Jawa Timur 60236, Indonesia
Lat -7.332831° Long 112.740469°
Friday, 21/11/2025 04:18 PM GMT +07:00

THANK YOU

javadsipa

